Contest-10

Problem 1: First, Last and Count

List · Basic

Problem Statement

Write a program that reads n numbers and stores them in a list. Then print the first number in the list, the last number in the list, and how many numbers the list holds.

Input Format

n + 1 lines of input:

- Line 1: n — how many numbers follow
- Next n lines: one integer each

Output Format

Print three lines in the exact format:

```
First: <first number>
Last: <last number>
Count: <how many numbers>
```

Constraints

- $1 \leq n \leq 100$
- $-1000 \leq$ each number ≤ 1000

Sample Test Case 0

Sample Input	Sample Output
5 10 20 30 40 50	First: 10 Last: 50 Count: 5

Explanation: The list is [10, 20, 30, 40, 50]. Its first item is 10, its last is 50, and it holds 5 items.

Sample Test Case 1

Sample Input	Sample Output
3 4 9 2	First: 4 Last: 2 Count: 3

Explanation: The list is [4, 9, 2].

Sample Test Case 2

Sample Input	Sample Output
1 7	First: 7 Last: 7 Count: 1

Explanation: With one item, the first and the last are the same number.

Solution

```
n = int(input())
nums = []
for i in range(n):
    value = int(input())
    nums.append(value)
print("First:", nums[0])
print("Last:", nums[-1])
print("Count:", len(nums))
```

Problem 2: Sum and Average of a List

List · Easy

Problem Statement

Write a program that reads n numbers into a list. Then go through the list to add up all the numbers, and print the sum and the average.

Input Format

n + 1 lines of input:

- Line 1: n — how many numbers follow
- Next n lines: one integer each

Output Format

Print two lines in the exact format:

```
Sum: <sum>
Average: <average, to 2 decimal places>
```

Constraints

- $1 \leq n \leq 100$
- $-1000 \leq$ each number ≤ 1000

Sample Test Case 0

Sample Input	Sample Output
5 10 20 30 40 50	Sum: 150 Average: 30.00

Explanation: 10 + 20 + 30 + 40 + 50 = 150, and 150 ÷ 5 = 30.

Sample Test Case 1

Sample Input	Sample Output
4 1 2 3 4	Sum: 10 Average: 2.50

Explanation: 10 ÷ 4 = 2.5, printed with 2 decimal places.

Sample Test Case 2

Sample Input	Sample Output
2 -4 10	Sum: 6 Average: 3.00

Explanation: Numbers can be negative: -4 + 10 = 6.

Solution

```
n = int(input())
nums = []
for i in range(n):
    nums.append(int(input()))
total = 0
for value in nums:
    total += value
average = total / n
print("Sum:", total)
print(f"Average: {average:.2f}")
```

Problem 3: Largest Number and Its Position

Problem Statement

Write a program that reads n numbers into a list, finds the largest number, and also tells where it sits in the list. Positions start from 0, just like list indexes do.

If the largest number appears more than once, print the position of its first appearance.

Input Format

n + 1 lines of input:

- Line 1: n — how many numbers follow
- Next n lines: one integer each

Output Format

Print two lines in the exact format:

```
Largest: <largest number>
Position: <its position in the list>
```

Constraints

- $1 \leq n \leq 100$
- $-1000 \leq$ each number ≤ 1000

Sample Test Case 0

Sample Input	Sample Output
5 3 7 2 9 4	Largest: 9 Position: 3

Explanation: The list is [3, 7, 2, 9, 4]. The largest is 9, at position 3 (counting from 0).

Sample Test Case 1

Sample Input	Sample Output
4 5 8 8 1	Largest: 8 Position: 1

Explanation: 8 appears at positions 1 and 2, so the first position, 1, is printed.

Sample Test Case 2

Sample Input	Sample Output
1 6	Largest: 6 Position: 0

Explanation: A list with one item has its largest item at position 0.

Solution

```
n = int(input())
nums = []
for i in range(n):
    nums.append(int(input()))
largest = nums[0]
position = 0
for i in range(n):
    if nums[i] > largest:
        largest = nums[i]
        position = i
print("Largest:", largest)
print("Position:", position)
```